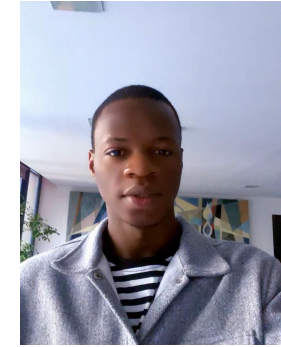


André Cédric Bessala - PEPR NF-NAI

Adaptive Energy Efficient RAN Function Placement in O-RAN



PhD Candidate

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IRIT-Toulouse INP/ CNRS

Gentian Jakllari & Guilherme Iecker Ricardo

Starting on October 11, 2023, with a duration of three years.

Motivation

- Mobile networks must drastically reduce their energy consumption.
- Current O-RAN control strategies are too rigid and waste energy under dynamic traffic.
- We exploit **PRB blanking** (intelligent radio sleeping) to unlock new energy-saving opportunities.

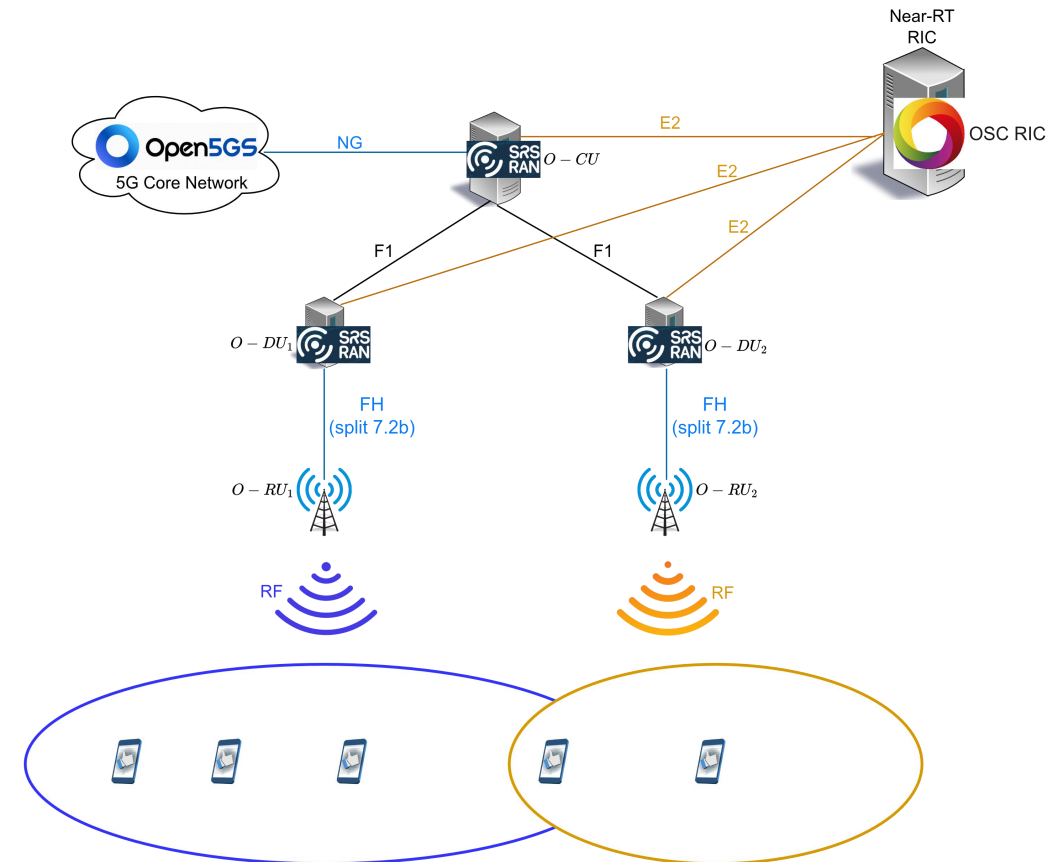
Contribution

- A distributed RL-based orchestration framework that jointly optimizes: user association, RAN function placement, and radio resource usage to **minimize energy consumption** while **maximizing operator profit**.

Ongoing Work

- We are developing on **Slices-RI testbed**, a learning framework that bridges the sim-to-real gap, enabling safe deployment of RL agents in real O-RAN networks.

UE-RU association & RAN function placement



PhD completion: Fall 2026, seeking academic and industrial positions
in AI-driven network optimization (from Jan 2027)